

ESP32-S3 DuckyScript Interpreter (CircuitPython)

Libraries Required

Download the **Adafruit CircuitPython Library Bundle** for your CircuitPython version: <https://circuitpython.org/libraries>

Copy these folders to the `lib/` directory on your ESP32-S3:

```
lib/
├─ adafruit_hid/
│   ├─ __init__.mpy
│   ├─ keyboard.mpy
│   ├─ keyboard_layout_us.mpy
│   ├─ keycode.mpy
│   ├─ mouse.mpy
│   └─ ...
└─
```

Minimum required: `adafruit_hid` bundle folder

Installation Steps

1. Flash CircuitPython on ESP32-S3

- Download UF2 for ESP32-S3 from: <https://circuitpython.org/downloads>
- Hold **BOOT** button, plug in USB, release BOOT
- Copy UF2 file to the USB drive that appears
- After reboot, CIRCUITPY drive appears

2. Copy Files

```
CIRCUITPY/
├─ boot.py          (optional - for stealth)
├─ code.py          <-- main DuckyScript interpreter
├─ payload.txt      <-- your DuckyScript payload
└─ lib/
    └─ adafruit_hid/
        └─ ...
```

3. Edit payload.txt

Write your DuckyScript commands. See supported commands below.

4. Run

- Plug ESP32-S3 into target PC
- Wait for PC to recognize it as a keyboard (2-3 seconds)
- Press the **BOOT** button (GPIO0)
- LED blinks = payload executed!

Supported DuckyScript Commands

Command	Example	Description
REM	REM this is a comment	Comment/ignored
DELAY	DELAY 1000	Wait N milliseconds
DEFAULT_DELAY	DEFAULT_DELAY 500	Delay after every line
STRING	STRING hello world	Type text literally
STRINGLN	STRINGLN hello	Type text + press ENTER
ENTER	ENTER	Press Enter key
SPACE	SPACE	Press Space
TAB	TAB	Press Tab
GUI / WINDOWS	GUI r	Windows/Super key
CTRL / CONTROL	CTRL c	Control key
ALT	ALT F4	Alt key
SHIFT	SHIFT INSERT	Shift key
UP / UPARROW	UP	Arrow Up
DOWN / DOWNARROW	DOWN	Arrow Down
LEFT / LEFTARROW	LEFT	Arrow Left
RIGHT / RIGHTARROW	RIGHT	Arrow Right
F1 - F12	F5	Function keys
ESC / ESCAPE	ESC	Escape key
BACKSPACE	BACKSPACE	Backspace
DELETE	DELETE	Delete key
HOME	HOME	Home key
END	END	End key
PAGEUP	PAGEUP	Page Up
PAGEDOWN	PAGEDOWN	Page Down
CAPSLOCK	CAPSLOCK	Caps Lock
PRINTSCREEN	PRINTSCREEN	Print Screen
SCROLLLOCK	SCROLLLOCK	Scroll Lock
PAUSE / BREAK	PAUSE	Pause/Break
NUMLOCK	NUMLOCK	Num Lock
MENU / APP	MENU	Application/Menu key
MINUS	MINUS	- key
EQUAL / EQUALS	EQUAL	= key
LEFTBRACKET	LEFTBRACKET	[key
RIGHTBRACKET	RIGHTBRACKET	] key
BACKSLASH	BACKSLASH	key
SEMICOLON	SEMICOLON	; key
QUOTE / APOSTROPHE	QUOTE	' key
GRAVE	GRAVE	key COMMA COMMA , key PERIOD PERIOD . key SLASH SLASH / key REPEAT REPEAT 5'

Example Payloads

Open Run and launch Notepad

```
DELAY 1000
GUI r
DELAY 500
STRING notepad
ENTER
DELAY 1000
STRING Hello from ESP32-S3!
```

Open CMD and run commands

```
DELAY 1000
GUI r
DELAY 500
STRING cmd
ENTER
DELAY 800
STRING whoami
ENTER
DELAY 500
STRING exit
ENTER
```

Open PowerShell (admin trick)

```
DELAY 1000
GUI r
DELAY 500
STRING powershell
CTRL SHIFT ENTER
DELAY 1000
LEFT
DELAY 200
ENTER
```

Rick Roll (Windows)

```
DELAY 1000
GUI r
DELAY 500
STRING https://www.youtube.com/watch?v=dQw4w9WgXcQ
ENTER
```

Create a file on Desktop

```
DELAY 1000
GUI r
DELAY 500
```

```
STRING notepad
ENTER
DELAY 1000
STRING You have been hacked!
CTRL s
DELAY 500
STRING %USERPROFILE%\Desktop\hacked.txt
ENTER
ALT F4
```

Stealth Mode (Optional)

In `boot.py`, uncomment these lines:

```
import storage
import usb_cdc

storage.disable_usb_drive() # Hides CIRCUITPY drive
usb_cdc.disable()           # Disables serial console
```

WARNING: After enabling stealth, you can only update code by:

1. Hold BOOT button while plugging in
2. Flash CircuitPython UF2 again (resets everything)
3. Or use a special “safe mode” button combo

Wiring / Pinout

Function	Pin	Note
BOOT Button	GPIO0	Built-in on DevKit
LED	GPIO38 or board.LED	Auto-detected
USB to PC	Native USB	Already connected

No external wiring needed! Just plug USB-C to PC.

Troubleshooting

Problem	Solution
“payload.txt missing!”	Create <code>/payload.txt</code> on CIRCUITPY drive
Nothing happens on PC	Wait 2-3 seconds after plugging in for HID enumeration
Wrong characters typed	This uses US keyboard layout. For other layouts, modify <code>KeyboardLayoutUS</code>
LED not blinking	Normal on some boards without built-in LED. Check serial console instead
Button doesn’ t work	Make sure you’ re pressing BOOT (GPIO0), not RESET

References

- CircuitPython HID: <https://docs.circuitpython.org/projects/hid/en/latest/>
- DuckyScript original: <https://docs.hak5.org/hc/en-us/articles/360010555153-Ducky-Script-the-USB-Rubber-Ducky-language>
- Adafruit HID Library: https://github.com/adafruit/Adafruit_CircuitPython_HID